Name: _____

Class: _____

Date: _____

Mr. Rawson • GCSE Physics

3.6 Gas Pressure and Temperature

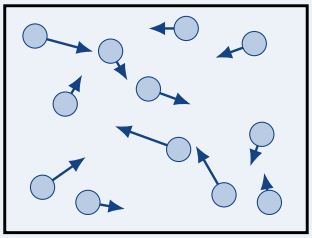
AQA Specification • §P3.3.1 Particle motion in gases • *P3.3 Particle model and pressure*

aP3-20 The molecules of a gas are in constant random motion. The temperature of the gas is related to the average kinetic energy of the molecules.
aP3-21 Changing the temperature of a gas, held at constant volume, changes the pressure exerted by the gas.
aP3-22 Students should be able to: explain how the motion of the molecules in a gas is related to both its temperature and its pressure; explain qualitatively the relation between the temperature of a gas and its pressure at constant volume.

Three ideas, each built on the one before: how the molecules move, how that makes a pressure, and what heating does to it.

1 · The molecules of a gas constant random motion

Key concept — motion and temperature



A longer arrow is a faster molecule.

The **molecules** of a gas are in **constant random motion**: they move all the time, in every direction, at a range of speeds. *Molecules* and *particles* mean the same thing here; the exam says molecules.

Some molecules are moving fast and some slowly, and they change speed every time they collide. So the gas has an **average** kinetic energy.

The temperature of a gas is related to the average kinetic energy of its molecules. Heat the gas and the molecules move faster *on average*, so the average kinetic energy rises, and that is what a higher temperature means.

1) **State** two things about the motion of the molecules in a gas. **[2]**

They are in constant motion (always moving); the motion is random, in all directions and at a range of speeds.

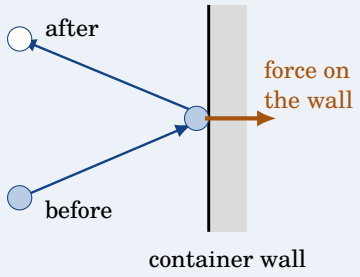
2) **Compare** the motion of the molecules in two sealed containers of the same gas, one at 20 °C and the other at 80 °C. **[2]**

In both, the molecules move randomly in all directions. At 80 °C the molecules have a higher average kinetic energy, so they move faster on average than at 20 °C.

Check: your answer to question 2 has the word **average** in it. A few molecules in the cold gas are faster than a few in the hot gas; it is the *average* kinetic energy that goes with the temperature.

2 · Where gas pressure comes from — collisions with the walls

Key concept — pressure



A moving molecule that reaches the wall of its container **collides** with it and bounces back. Each collision pushes on the wall: a tiny **force**, for a tiny moment.

There are millions of millions of these collisions on every part of the wall every second, so the wall feels a **steady force**. Spread over the area of the wall, that force is the **pressure** of the gas.

Say it in three links: collisions with the wall → a force on the wall → a pressure on the wall.

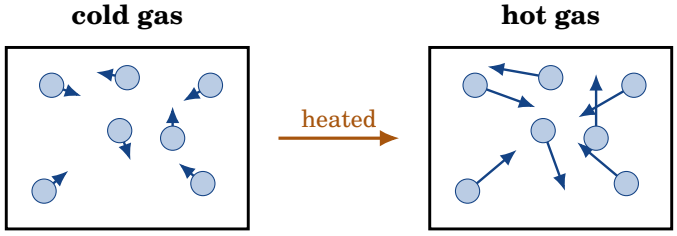
3) **Explain** how the molecules of a gas exert a pressure on the walls of their container. **[3]**

The molecules are moving randomly, so they collide with the walls. Each collision exerts a force on the wall. Millions of collisions every second give a steady force over the whole area of the wall, and this force over the area is the pressure.

Check: your answer uses **collide**, **force** and **pressure**, in that order. Each word is caused by the one before it.

3 · Heating a gas at constant volume — faster molecules, more pressure

A **rigid, sealed** container keeps the volume of the gas the same and keeps every molecule inside. Heat it, and only the *motion* of the molecules changes.



Same container, same volume, same number of molecules. Only the arrows have changed.

Worked example — copy this shape

Explain why the pressure of a gas in a sealed, rigid container increases when the gas is heated.

1. Heating the gas gives its molecules more kinetic energy, so they move **faster on average**.
2. Faster molecules reach the walls sooner, so they collide with the walls **more often**.
3. Each collision is harder, so it exerts **a larger force** on the wall.
4. The volume is fixed, so the same wall area takes more collisions, each with more force: the **pressure increases**.

Check: a full answer has all four links, in order: faster → more often → more force → pressure up. To explain *cooling*, write the same four links with each one reversed.

4) **Explain** what happens to the pressure of the gas in a sealed, rigid container when the gas is **cooled**. Use the four links. **[4]**

Cooling the gas lowers the kinetic energy of its molecules, so they move more slowly on average. Slower molecules collide with the walls less often, and each collision exerts a smaller force on the wall. The volume is fixed, so fewer collisions with less force on the same area means the pressure decreases.

4 · The same chain in the real world rigid containers, sealed

5) **Explain** why a car tyre's pressure gauge reads higher after a long, fast drive than it did before the car set off. Treat the tyre as a rigid, sealed container. **[3]**

The drive warms the tyre, so the air inside is heated. The molecules move faster on average, collide with the inside of the tyre more often and with more force. The volume is fixed, so the pressure increases and the gauge reads higher.

6) **Explain** why the lid of a jar that was sealed while hot is harder to open once the jar has cooled in the fridge. The air outside the jar stays at the same pressure. **[3]**

Cooling the air inside the jar makes its molecules move more slowly on average, so they collide with the lid less often and with less force: the pressure inside the jar decreases. The air outside is still at the same pressure, so it pushes the lid onto the jar with a larger force than the air inside pushes it off, and the lid is harder to open.

Exam-style question [4 marks]

An aerosol can is a rigid, sealed metal container holding gas under pressure. Every can carries this hazard label: *"Pressurised container. Protect from sunlight. Do not expose to temperatures exceeding* 50 °C*."*

Use the particle model to explain why heating the can is dangerous.

Heating the can heats the gas inside, so the molecules gain kinetic energy and move faster on average. They collide with the walls of the can more often, and each collision exerts a larger force. The can is rigid, so the volume stays the same and the pressure of the gas increases. If the pressure becomes large enough, the force on the walls is greater than the can is built to hold and the can bursts. 1 mark each: faster on average; more frequent collisions; more force per collision; pressure rises at fixed volume so the can may burst.